

Finite Abelian

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Problem 0.

Proof. $\forall r \in S, r * 0_s + r = r * (0 + 1_s) = r(1_s) = r \implies r * 0_s = 0.$

$\forall r \in S, 0 = r * 0_s = r * 1_s = r$, thus $r = 0$. Thus $S = \{0\}$ □

Problem 1.

Proof. If S acts on A , it means there $\exists f : S \times A \rightarrow A$ with the following properties, $\forall r, s \in S$ and $\forall a, b \in A$

1. Distributivity over group addition: $f(r, a + b) = f(r, a) + f(r, b)$,
2. Distributivity over ring addition: $f(r + s, a) = f(r, a) + f(s, a)$
3. Compatibility over ring multiplication: $f(rs, a) = f(r, f(s, a))$
4. Multiplicative identity element: $f(1, a) = a$

Let $g : S \rightarrow \text{Hom}_{gr}(A, A)$, a ring homo. Where $\forall s \in S, g(s) := [a \rightarrow f(s, a)]$. Thus g determines the choice of f and vice versa.

Let $g_1, \dots, g_k$, be the generators of M . $\forall m \in M, m = r_1 g_1 + \dots, r_k g_k$. Thus the map, sending $(r_1, \dots, r_k) \rightarrow m$ is surjective.

A determines M , because the number of rows determine the number of generators of M , while each column describes one of the relations which describe M . Since generators and relations determine a group, A determines a group with the same generators and relations as M , and thus isomorphic (in this case equal) to M , by the universal property. □

Problem 2.

Proof. $M(A) \cong S^g / \text{Im}(A)$ and $M(RAC) \cong S^g / \text{Im}(RAC)$.

$\text{Im}(AC) \cong \text{Im}(A)$, because C is an isomorphism of S^k , thus the image doesn't change.

$\forall x + \text{Im}(RAC) \in M(RAC), \exists y + \text{Im}(A) \in M(A) \ni R_*(y + \text{Im}(A)) = x + \text{Im}(RAC)$, because R is invertible and C is invertible and thus isomorphisms. Thus R_* is a surjective map from $M(A)$ to $M(RAC)$.

$\forall a + \text{Im}(A), b + \text{Im}(A) \in M(A) \ni R_*(a + \text{Im}(A)) = R_*(b + \text{Im}(A)) \implies Ra + R_*(\text{Im}(A)) = Ra + R_*(\text{Im}(AC)) = Ra + \text{Im}(RAC) = Rb + R_*(\text{Im}(B)) = Rb + R_*(\text{Im}(AC)) = Rb + \text{Im}(RAC) \implies a + \text{Im}(AC) = b + \text{Im}(AC)$, because R is invertible. Thus R_* is injective.

$\forall a + \text{Im}(A), b + \text{Im}(A), \forall r, s \in S, R_*(r(a + \text{Im}(A)) + s(b + \text{Im}(A))) = R_*((ra + sb) + \text{Im}(A)) = ra + sb + \text{Im}(RAC) = ra + \text{Im}(RAC) + sb + \text{Im}(RAC) = rR_*(a + \text{Im}(A)) + sR_*(b + \text{Im}(A))$

Thus $M(A) \cong M(RAC)$ □

Problem 3.

Proof. Since $M(RAC) \cong M(A)$, without loss of generality, let A be "diagonal" after row and column operations. Thus $A = \begin{bmatrix} D & 0 \\ 0 & 0 \end{bmatrix}$ where $D = \begin{bmatrix} a_1 & & \\ & \ddots & \\ & & a_n \end{bmatrix}$. Let $a_i = 0 \forall i > n$.

Thus $\forall v \in S^k$, $v = (s_1, \dots, s_k)$ and $Av = (a_1 s_1, \dots, a_n s_n, 0, \dots, 0)$. Thus $\forall v \in A_*(S^k)$, the i th component of v are multiples of a_i . Thus $Im(S^k) \leq \bigoplus_{i=1}^n a_i S$, however $\forall (s_1, \dots, s_n) \in \bigoplus_{i=1}^n a_i S$, $(\frac{s_1}{a_1}, \dots, \frac{s_n}{a_n}, 0, \dots, 0) \in S^k$, $A(\frac{s_1}{a_1}, \dots, \frac{s_n}{a_n}, 0, \dots, 0) = (s_1, \dots, s_n, 0, \dots, 0) \implies (s_1, \dots, s_n, 0, \dots, 0) \in A_*(S^k) \implies (s_1, \dots, s_n) \in A_*(S^k)$, naturally. Thus, $Im(S^k) \cong \bigoplus_{i=1}^n a_i S$

Consider the map, $\phi : M(A) \rightarrow \bigoplus_{i=1}^n S/a_i S$ which sends the equivalence class of $(x_1, \dots, x_g) \rightarrow (x_1 + a_1 S_1, \dots, x_n + a_n S_n, \dots, x_g)$

$\forall w \in A_*(S^k)$, by the above, $w = (a_1 s_1, \dots, a_n s_n, 0, \dots, 0)$, $\phi((x_1, \dots, x_g) + w) = \phi(x_1 + a_1 s_1, \dots, x_n + a_n s_n, x_{n+1}, \dots, x_g) = (x_1 + a_1 S_1, \dots, x_n + a_n S_n, \dots, x_g) = \phi((x_1, \dots, x_g))$, so well defined

$\forall (x_1, \dots, x_g), (y_1, \dots, y_g) \in S^g \ni \phi((x_1, \dots, x_g)) = \phi((y_1, \dots, y_g)) \implies x_i - y_i \in a_i S_i \forall i \implies x - y \in A_*(S^k) \implies (x_1, \dots, x_g) = (y_1, \dots, y_g)$ in $M(A)$

$\forall y = (a_1 s_1 + a_i S, \dots, a_n s_n + a_i S) \in \bigoplus_{i=1}^n S/a_i S$, $\phi(a_1 s_1, \dots, a_n s_n, 0, \dots, 0) = y$.

Thus $M(A) \cong \bigoplus_{i=1}^n S/a_i S$ □

Problem 4.

Proof. $\forall m, n \in \mathbb{N}$.

Inductive step: Suppose all $k \times l$ where $k < m$ and $l < n$ matrices satisfy the theorem.

Inductive Hypothesis: Let A be a matrix of size $m \times n$. Let R and C be the necessary row and column permutation matrices to get the smallest possible entry of A to the top right corner of matrix, ie the R and C such that $[RAC]_{1,1}$ has the smallest nonzero absolute value.

For each row r_i in A , by the Division algorithm, $\exists q, r \in \mathbb{Z}$ For each row r_i in A , by the Division Algorithm, $\exists q, r \in \mathbb{Z} \ni q a_{1,1} + r = a_{i,1}$. Multiply R by the row operation which replaces the r_i with $r_i - q r_1$. Now $a_{i,1} = r$. But since the choice of R and C ensured that $a_{i,i}$ is the smallest possible nonzero value and $r < a_{i,i}$, thus $r = 0$. Thus $a_{1,1}$ divided the elements of the column. Similarly for the columns, $r = 0$ and the elements of the first row are divisible by $a_{1,1}$.

If $m = 1$ or $n = 1$, then we are done.

If $m, n \geq 2$, we can get the j th row to the first row with the row operation $r_1 + r_i \rightarrow r_1$ to get the matrix A' . For each element of the first row of A' , by the euclidean algoirthm, $\exists q, r \in \mathbb{Z} \ni q a'_{1,1} + r = a'_{1,i}$. Do the column operation $c_i - q c_1 \rightarrow c_i$. Now $a'_{1,i} = r$. By the choice of R and C has ensured that $a'_{1,1}$ is the smallest possible nonzero value, thus $a_{1,1} | a'_{1,i}$ thus $a_{1,1} | a_{j,i}$. Thus $a_{i,i}$ divides every entry of A .

Thus submatrix B excluding the first column and row is $a_{1,1} B'$.

Thus by the induction hypothesis, B can diagonalized where $b_{i,i} | b_{i+1,i+1}$. Since $B = a_{1,1} B'$, thus $a_{1,1} | b_{i,i}$.

Thus by induction any matrix can be diagonal using row and column operations where i -th diagonal entry divides the $i + 1$ diagonal entry. □

Problem 5.

Proof. Let G be a finitely generated abelian group. Thus by problem 1 and 2, there exists a finite presentation of G which is the matrix A in the ring of integers. Thus by definition, $M(A) = G$. By 3, $M(A) \cong \bigoplus_i^n (\mathbb{Z}/a_i\mathbb{Z}) \oplus \mathbb{Z}^r$ (as vectorspaces). By 4, we can pick the a_i such that $a_1 | \dots | a_n$. $\square$